Extra Dessert #11: Propagation Prediction and Space Weather

A-index, K-index, Bz, solar flare classes, geomagnetic storms, radio horizon, and VOACAP.

Group E3C · 12 pool questions · Self-study reference · Practice exams at eham.net/exams

Solar Activity And Its Effects

Question	Answer	Q
Cause of short-term radio blackouts	**Solar flares**	E3C01
Greatest solar flare intensity class	**Class X** (A, B, C, M, X – X is highest)	E3C07
What 304A measures	**UV emissions at 304 angstroms, correlated to the solar flux index**	E3C10
Sudden HF noise rise across the band	**A coronal mass ejection impact or a solar flare has occurred**	E3C12

Geomagnetic Indices

Question	Answer	Q
Rising A-index or K-index indicates	**Increasing disturbance of the geomagnetic field**	E3C02
Path most absorbed when A/K is elevated	**Through the auroral oval**	E3C03
Term for an extreme geomagnetic storm	**G5** (G-scale ranges G1-G5; G5 = extreme)	E3C08

The Bz Parameter

Question	Answer	Q
What Bz represents	**North-south strength of the interplanetary magnetic field**	E3C04
Bz orientation that increases disturbance risk	**Southward** (a southward IMF opens the magnetosphere to solar particles)	E3C05

Radio Horizon And Prediction Tools

Question	Answer	Q
VHF/UHF radio horizon vs geographic horizon	**Approximately 15% farther** (atmospheric refraction bends radio waves slightly)	E3C06
What VOACAP models	**HF propagation**	E3C11
What amateur propagation reporting networks report	**Digital-mode and CW signals** (e.g., PSK Reporter, WSPRnet, RBN)	E3C09

Dessert 11 Question Bank (E3C)

E3C01. What is the cause of short-term radio blackouts?
A. Coronal mass ejections.
B. Sunspots on the solar equator.
C. North-oriented interplanetary magnetic field.
D. Solar flares.

E3C02. What is indicated by a rising A-index or K-index?
A. Increasing disturbance of the geomagnetic field.
B. Decreasing disturbance of the geomagnetic field.
C. Higher levels of solar UV radiation.
D. An increase in the critical frequency.

E3C03. Which of the following signal paths is most likely to experience high levels of absorption when the A-index or K-index is elevated?
A. Transequatorial.
B. Through the auroral oval.
C. Sporadic-E.
D. NVIS.

E3C04. What does the value of Bz (B sub z) represent?
A. Geomagnetic field stability.
B. Critical frequency for vertical transmissions.
C. North-south strength of the interplanetary magnetic field.
D. Duration of long-delayed echoes.

E3C05. What orientation of Bz (B sub z) increases the likelihood that charged particles from the Sun will cause disturbed conditions?
A. Southward.
B. Northward.
C. Eastward.
D. Westward.

E3C06. How does the VHF/UHF radio horizon compare to the geographic horizon?
A. It is approximately 15 percent farther.
B. It is approximately 20 percent nearer.
C. It is approximately 50 percent farther.
D. They are approximately the same.

E3C07. Which of the following indicates the greatest solar flare intensity?
A. Class A.
B. Class Z.
C. Class M.
D. Class X.

E3C08. Which of the following is the space-weather term for an extreme geomagnetic storm?
A. B9.
B. X5.
C. M9.
D. G5.

E3C09. What type of data is reported by amateur radio propagation reporting networks?
A. Solar flux.
B. Electric field intensity.
C. Magnetic declination.
D. Digital-mode and CW signals.

E3C10. What does the 304A solar parameter measure?
A. The ratio of X-ray flux to radio flux, correlated to sunspot number.
B. UV emissions at 304 angstroms, correlated to the solar flux index.
C. The solar wind velocity at an angle of 304 degrees from the solar equator, correlated to geomagnetic storms.
D. The solar emission at 304 GHz, correlated to X-ray flare levels.

E3C11. What does VOACAP software model?
A. AC voltage and impedance.
B. VHF radio propagation.
C. HF propagation.
D. AC current and impedance.

E3C12. Which of the following is indicated by a sudden rise in radio background noise across a large portion of the HF spectrum?
A. A temperature inversion has occurred.
B. A coronal mass ejection impact or a solar flare has occurred.
C. Transequatorial propagation on 6 meters is likely.
D. Long-path propagation on the higher HF bands is likely.

Answer Key

E3C01: D	E3C02: A	E3C03: B	E3C04: C
E3C05: A	E3C06: A	E3C07: D	E3C08: D
E3C09: D	E3C10: B	E3C11: C	E3C12: B